NYAMBARIA BOYS' 2025 PREDICTION

KCSE PROJECTION EXAM

COMPUTER STUDIES

451/2

PAPER 2

TIME: 2½ HOURS

NAME………………………………………………………………………………………

SCHOOL……………………………………………… SIGN………………………

INDEX NO…………………………………………… ADM NO……………………

Kenya Certificate of Secondary Education

INSTRUCTIONS TO CANDIDATES

a) *Indicate your name and index number at the right hand corner of each printout*

b) *Write your name and index number on the CD/removable storage medium provided*

c) *Write the name and version of the software used for each question attempted in the answer sheet provided*

d) *Answer all the questions, All questions carry equal marks*

e) *Passwords should not be used while saving in the CD/removable storage Medium*

f) *Marked printout of the answers on the sheet*

g) *Hand in all the printouts and the CD/removable storage medium used*

For Marking Schemes   Contact 0746 222 000 / 0742 999 000

1. (a)(i) The extract below shows a spreadsheet used to compute the toll charges for a highway based on the type of vehicle, tonnage and charge per kilometer for usage.

	HIGHWAY TOLL CHARGES					
Registration	**Vehicle Type**	**Weight**	**Distance**	**NormalCharge**	**Penalty Charge**	***Total***
KCY 789 M	**PickUp**	**6**	**12**			
KCR 769 L	**Car**	**4**	**40**			
KCF 724 C	**PickUp**	**6**	**32**			
KCM 737 N	**Truck**	**12**	**25**			
KCA 745 W	**Lorry**	**20**	**28**			
KCP 756 H	**Truck**	**10**	**12**			
KCU 778 J	**Car**	**4**	**8**			
KCZ 701 A	**PickUp**	**8**	**25**			
KCB 781 E	**Car**	**6**	**4**			
KCV 743 H	**PickUp**	**4**	**20**			
KCQ 735 X	**Truck**	**8**	**32**			
KCT 721 K	**Lorry**	**10**	**25**			
KCD 792 V	**Truck**	**12**	**28**			
KCZ 784 P	**Car**	**6**	**12**			
KCB 756 C	**Truck**	**10**	**8**			
KCE 734 D	**Car**	**4**	**25**			
KCF779 E	**PickUp**	**6**	**32**			
KCG 700 F	**Lorry**	**12**	**25**			
KCH 723 K	**Truck**	**20**	**28**			
KCJ 711 W	**PickUp**	**10**	**12**			
KCR 712 D	**Car**	**4**	**8**			
KCD 774 B	**PickUp**	**8**	**25**			
KCS 756 M	**Truck**	**6**	**4**			

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KCA 745 W	**Car**	**4**	**20**			
			Total			

(ii) Create a workbook and save the workbook as toll. **(2 Marks)**

(iii)Fill the data in the worksheet1 and rename the worksheet as tollOriginal.**(14 Marks)**

(b)(i)The NormalCharge column is calculated based on the table below.

Vehicle Type	Max AllowedWeight(Tns)
PickUp	6
Car	4
Truck	8
Lorry	10

(ii) The PenaltyCharge column is calculated based on the table below. The penalty is based on any weight above Maximum allowed weight for a vehicle type for every kilometer of the the usage. **(8 Marks)**

Vehicle Type	Penalty Charge (Ksh)per Km
PickUp	10
Car	5
Truck	15
Lorry	20

(iii) The TotalCharge is based on summation of NormalCharge and PenultyCharge . Create a column TotalCharge and use a function to Calculate the Total Charge **(2 Marks)**

(iv) Create the Running Totals for Normal Charge,Penulty Charge And Totalcharge **(4 Marks)**

(b)(i) Copy The data in the OriginalToll to another worksheet rename the workshhet as Sorted**(1 Mark)**

(ii) Sort the Data is ascending order of Vehicle type Sorted worksheet. **(4 Marks)**

(**iii)**Create subtotals based on the vehicle Type **(4 Marks)**

(iv) Draw a column chart based on The Vehicle Type subtotals and Total Charge (**8 Marks)**

2. a) Create a database called **Aberdare bottles ltd** and create the following tables **(15 marks)**

Table 1: Employee

Employee_ID	EmployeeName	Department	YearOfEmployment
101	Kibet Arap Kamau	Human resource	1985

102	Janet Atieno	Procurement	1990
450	Kimani Koigu	Accounts	2000
891	Moraa Kerubo	Human resource	2010

Table 2: Sales

ProductName	**Employee_ID**	**ProductID**	**SalesAmount**	**Salary**
Tea leaves	101	Xc101	5000	
cocoa	102	Xp105	15500	
coffee	450	Xvb11	9500	
Chocolate	891	X56po	30000	

Table 3: Department

Employee_ID	**Department_Name**	**HeadOfDepartment**	**NoOfEmployees**
101	Human resource	B.N. Komu	52
102	Procurement	J.K. Wanjiru	12
450	Accounts	P.G. Otindo	20
891	Human resource	M.M. Jerotich	10

i) Create **relationship** among the tables **(2 marks)**

ii) Create **three** input screens **(forms)** and use them to enter the data into the tables above**(6 marks)**

iii) Create a query called **Start_K** and use it to display **EmployeeName** that start with letter **K** **(3 marks)**

iv) Display the **no of years** an employee has worked given that the current year is 2018. Save **the report** as **AGE.** **(3 marks)**

v) Create a query called **Yote** to display the following fields **(2 marks)**

- **Employee_ID**
- **EmployeeName**
- **Department_Name**
- **ProductName**
- **Salary**
- **HeadOfDepartment**

vi) Copy Yote query (**in v above**) and save the new query as **MPYA**:- **(1 marks)**

Use MPYA query to

✓ Calculate the salary given that: salary is 10% of the S**alesAmount** **(2 marks)**

✓ Display salary in ascending order **(2 marks)**

✓ Display employees from **human resource** department whose **SalesAmount** is **greater than 12000.** **(2 marks)**

vii) Create a form called **AberdareForm** using **Yote query** (**in v above**) and use it to answer the questions below:-

✓ Count no of employees **(2 marks).**

✓ Add a title of the form as **"Aberdare bottles ltd-2018"** **(2 marks)**

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✓ Insert **date and time** on the form header use **(=NOW( ))** **(2 marks)**

viii) **Print** Age, Sales table, and AberdareForm **(3 marks)**